

HARSHIT PAL

Full-Stack Developer & AI / IoT Systems Engineer

Ghaziabad, UP, India | harshit.pal.8.d.sdpsmzn@gmail.com | harshitpal.in | [GitHub](https://github.com/harshitpal) | [LinkedIn](https://www.linkedin.com/in/harshitpal) | [@harshit_siraswal](https://twitter.com/harshit_siraswal)

HONORS, CERTIFICATIONS & NATIONAL QUALIFICATIONS

- **2x NDA Qualified & 3x SSB Conference Out:** Cleared National Defence Academy entrance exam twice and attended 3 Services Selection Boards (demonstrated national-level leadership, psychological resilience, and crisis decision-making).
- **IOQM 2022 Qualified:** Merit Certificate Holder in Indian Olympiad Qualifier in Mathematics (awarded for top national mathematical problem-solving talent).
- **AWS Certified AI Practitioner:** Industry-recognized certification for generative AI architectures, foundational models, cloud compliance, and ML pipeline development.
- **Innotech 2025 Hackathon:** First-Year Innovator recognition at KIET Ghaziabad for engineering functional IoT hardware mesh systems.
- **JEE Main 2025:** 89.12 Percentile in national level engineering entrance examination.

EDUCATION

KIET Group of Institutions — <i>B.Tech in CSE (Artificial Intelligence & Machine Learning)</i>	2025 – 2029 CGPA: 7.84 (Sem 1: 7.83, Sem 2: 7.86)
SD Public School — <i>Senior Secondary (Class XII: 81.2%) & High School (Class X: 94.6%)</i>	Muzaffarnagar, UP

TECHNICAL PROJECTS

BIS Standards Recommendation Engine AI / Hybrid RAG Retrieval Engine	Live Demo GitHub
• Sub-0.01s Hybrid Search: Engineered deterministic RAG system parsing Bureau of Indian Standards (BIS SP 21) cement/steel specifications for Indian MSEs using dense FAISS vector embeddings and sparse BM25 retrieval. • Benchmark Precision: Achieved 100% Hit Rate @3, MRR @5 of 1.000, and automated validation passing 59/59 robustness tests with FastAPI and domain term expansions.	
StudyShare Suite & Ingestion Automation Next.js, Flutter, NestJS, Neon, Tesseract, n8n	studyshare.in GitHub
• Campus ERP Ecosystem: Built live campus portal and mobile app deployed on Railway/Neon PostgreSQL serving 500+ active users for leave requests, fee verification, and complaint dispatch. • Automated OCR Ingestion: Shipped an n8n webhook pipeline integrating Tesseract OCR for automated fee receipt parsing (98.4% accuracy, 1.2s latency) and dynamic encrypted QR student gate pass verification.	
ReturnShield AI (Innogeeks Hackathon) Python, scikit-learn, React 19, Vite, Edge	returnsheild-ai.vercel.app GitHub
• ML Return Risk Classifier: Built an e-commerce fraud and return prevention platform at Innogeeks Hackathon using trained scikit-learn classifiers with real-time order risk scoring (95.0% accuracy, 45ms latency) and revenue impact estimation. • Interactive Merchant Intelligence: Implemented a high-performance React 19 + Geist design dashboard with automated action stacks, 12-week risk trend visualization, and Vercel Edge Copilot.	
ESP32 Smart Infrastructure & IoT Mesh Networks C++, ESP-NOW, MQTT, FreeRTOS	Wokwi Sim Demo
• Zero-GSM Urban Mesh: Developed low-power municipal manhole monitoring hardware utilizing ESP32, HX711 load cells, and ultrasonic sensors communicating over self-healing ESP-NOW peer-to-peer mesh relays.	
Code Analyser Platform React, FastAPI, Monaco Editor, Judge0, AST Parser	code-analyser-web.vercel.app GitHub
• Automated Code Evaluation: Created practice compiler sandbox utilizing Abstract Syntax Tree (AST) parsing and Judge0 execution pipeline to classify logic quality, syntax anomalies, and big-O computational complexity.	

TECHNICAL SKILLS & CORE COMPETENCIES

Languages:	TypeScript, JavaScript, Python, Dart, C++, C, SQL, HTML5/CSS3
Frontend & Mobile:	React, Next.js, Flutter, Tailwind CSS, shadcn/ui, Vite, Framer Motion
Backend & APIs:	Node.js, Express, NestJS, FastAPI (Python), Prisma ORM, RESTful APIs, WebSockets
AI, Cloud & Data:	FAISS Vector DB, Hybrid RAG, BM25, scikit-learn, PostgreSQL, Neon Serverless, Redis, Docker, Vercel, Railway, Cloudflare, Linux, n8n
Hardware & IoT:	ESP32, ESP-NOW Mesh Protocol, Arduino, MQTT, FreeRTOS, Sensor Interfacing (PIR, Ultrasonic, Load Cells)